Timber Technology

Land Sciences

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Timber Technology (A09842)

Short Title: Timber Technology
Faculty Science and Computing
Department Land Sciences
Cluster Forestry
Author MPEDINI
Credits 5 **Level:** Introductory

Description of Module / Aims

In this module students will deal with wood as an end product rather than as a raw material. Through factory visits students will be studying the main conversion processes from round wood to finished product. Students will also get the opportunity to explore the properties and limitations of wood as a material in conjunction with other materials (wood composites). Finally, the possibilities of improving timber properties through preservative treatment and wood modification will be explored.

Programmes

FORS-0006	BSc in Forestry (WD_SFORS_D)	2 / 3 / M

Indicative Content

- The market for wood-based products in Ireland
- Round wood grading and storage
- Round wood conversion processes in softwood and hardwood sawmills
- Timber drying
- Wood preservation and wood modification
- Engineered timber products incl. panel boards
- Wood combustion and combustion technology

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the Irish supply chain for wood-based products.
2. Outline the production flow in softwood mills, hardwood mills, and board mills using the correct terminology.
3. Name and explain the grading parameters for round wood and sawn timber and relate it to forest management decisions.
4. Describe the requirements for storage of round wood logs and sawn timber.
5. Define the key concepts of the wood/water relationships involved in timber drying.
6. Describe basic combustion technology and relate it to the developing wood energy market in Ireland.
7. Describe the possibilities for enhancing wood properties through timber preservation, engineered wood products and wood modification.
8. Compare the effectiveness of different strategies adopted by sawmills to maximise yield from round wood.

Learning and Teaching Methods

- Key concepts will be described in detail in lectures.

- A number of field trips will be organised to Irish wood industries, where the interrelation between the round wood material and the finished product will be explored along with current production technology.
- Students will be encouraged to independently study recommended reading and video clips provided on the module's Moodle page.

Assessment Methods

	Weighing		Outcomes Assessed
Final Written Examination	50%		4,5,6,7
Continuous Assessment	50%		
Assignment		35%	1,2,3,8
In-Class Assessment		15%	1,2,6,7

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/ skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out assessments. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design and current knowledge limitations when carrying out assessments. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on assessments. Show initiative, individual thought, and enthusiasm in self-study and independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Field Trips	24	
Independent Learning	87	

Essential Material

- Bowyer, J.L., R. Schmulsky and J.G. Haygreen. _Forest products and wood science : an introduction_. 5th ed.. Ames: Blackwell Publishing, 2007.
- IFFPA, . _An overview of the Irish forestry and forest products sector 2015_. Dublin: Irish Forestry and Forest Products Association, 2015.

Supplementary Material

- "Woodspec." www.woodspec.ie
- Davies, I. and G. Watt. _Making the Grade -- A guide to appearance grading UK grown hardwood timber._. Scotland: Forestry Commission Scotland, Arcamedia, 2005.
- Dinwoodie, J.M. _Timber, its nature and behaviour_. 2nd ed.. New York: Routledge, 2000.
- FPL, . _Wood Handbook---Wood as an engineering material_. Madison, WI: Dept. of Agriculture, Forest Services, Forest Products Laboratory, 2010.
- Thoemen, H., M. Irle and M. Sernek (eds). _Wood-Based Panels - An Introduction for Specialists_. London: Brunel University Press, 2010.

Requested Resources

- Lecture Room: Loose Seated